Islamic University of Technology

Lab 6: Iteration Statements

CSE 4108 : Structured Programming Lab

Section 2B

MIA, NFS, IAC, ASH

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1 Lab Tasks

1.1 Who's the Senior

A student ID of Foxford University (FoxU) is represented in the format `YY-D-SN` where `YY` is the last two digits of batch no, `D` is Department ID and `SN` is Serial No for that student.
Write a C program that continuously takes student IDs as input from the user and stops only when the user enters 0 as *YY* input (in this case, *D* and *SN* values do not matter).. Then the program displays the most senior student ID entered so far. The order of comparison for checking seniority is `lower batch number > higher dept. id > lower serial no`. The first input of the program will always be a valid ID.

[Sample Program]
Input 1

```
Enter an integer: 18-2-16
Enter an integer: 20-4-55
Enter an integer: 16-4-48
Enter an integer: 16-2-57
Enter an integer: 16-4-69
Enter an integer: 0-0-0
```

Output 1

```
The senior: 16-4-48
```

1.2 Only Diamond is Real

Mr. ChaCha, a great computer programmer, has come up with a plan of being the most wealthy man in the world using his programming skill. He wants to print `Diamonds` using a program as diamonds are rare and sell those diamonds to achieve his goal. Good for you, you've come to know of his plan through your surveillance that you were doing on him. Now, you'll print those diamonds before him and be the richest man in the world.

Write a program in C that takes an integer `r` as input and print the diamond with height `2r - 1` like the sample output. The minimum value of `r` would be 2 and max value would be 25.

[Sample Program]
Input 1

```
Enter number: 2
```

Output 3

```
 *
***
 *
```

Input 2

```
Enter number: 4
```

Output 2

```
    *
   ***
  *****
 *******
  *****
   ***
    *
```

[Hint] *Try to understand the pattern of printing, count based on the input value, how many spaces and how many asterisks are there in each line. Once you derive the relation, the printing should be easy. You need to use loop in loop to achieve the desired output. You might want to use separate loops for upper and bottom portion.*

1.3 Primal Range

Write a C program that would take two numbers (greater than or equal to 2) as input and print out all the prime numbers between those two number (inclusive) and the count of primes.

[Sample Program]
Input 1

```
Enter range (inclusive) : 10 60
```

Output 1

```
Primes between 10 and 60: 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47,
53, 59
Prime Count: 13
```

[Hint] *Utilize the prime example shown in theory class. You can use the slide. Loop the prime check for all the numbers in given range.*

1.4 Arc Tan

The following series formula can be used to approximate the value of π.

$$\frac{\pi}{2} = 1 + \frac{1}{3} + \frac{1*2}{3*5} + \frac{1*2*3}{3*5*7} + \frac{1*2*3*4}{3*5*7*9} + ...$$

Write a C program that would take the number of elements of the series to consider and output the approximate value of π. For greater accuracy, your program should use variables of type `double` rather than `float`. Have the program produce output to a precision of 8 decimal places.

[Sample Program]
Input 1

```
Enter number of elements: 3
```

Output 1

```
Pi: 2.93333333
```

1.5 Serial Calculator

Write a program that evaluates an expression. The operands in the expression are floating-point numbers; the operators are +, -, *, and /. The expression is evaluated from left to right (no operator takes precedence over any other operator).

[Sample Program]
Input

```
Enter an expression: 1+2.5*3
```

Output

```
Value of expression: 10.5
```

[Hint] *Alternatively scan floating point numbers and characters. The newline will be a character.*

1.6 Fractured Again

In *Formatted I/O Lab*, you were asked to write a program that would add or subtract two fractions directly using mathematics. Now use character to implement a C program which would take two fractions as input and print out their summation, subtraction, product or quotient based on the operator (+, -, *, /) in the middle. The numerator and denominator of the fractions can be in trillion range (999,999,999,999,999). Numerator and denominator are integral type and `0` will not be given as input for the denominators of the fractions. No need to express the resultant fraction in lowest/reduced form.

[Sample Program]
Input

```
Enter a fraction: 999999999999/999999999999 +  1/2
```

Output

```
Result: 2999999999997/1999999999998
```

[Hint] *Type* `int` *might not be enough for this problem. Look for something larger.*